

Cross-Platform Dual Node Synchronization & Peer Connectivity Validation

Windows & Linux (WSL) Post-Update Infrastructure Verification

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Report Type	Infrastructure Validation — Node Synchronization & Peer Connectivity
Relates to	Reports 1–12 of the DAC Testnet (Inception) Technical Report Series
Report Archive	https://github.com/EdLWEISS186/dac-dual-node-cgnat-setup/tree/main/Testnet (Inception) Technical Reports

Executive Summary

This report documents a follow-up infrastructure validation cycle performed on the DAC Testnet dual-node single machine setup after both nodes were updated to the latest client release (gdacnode v1.11.6). The objective was to confirm that the Windows and Linux (WSL) deployments continued to synchronize correctly, maintain healthy peer connectivity, and remain compatible with one another and with official DAC infrastructure following the update. Both nodes completed synchronization successfully, mutually discovered each other as active peers, and maintained stable connections to official DAC Authority and RPC nodes. No abnormal synchronization behaviour or network instability was observed.

Component	Status	Detail
Windows Node Sync	Confirmed	eth.syncing = false, 17 peers
Linux (WSL) Node Sync	Confirmed	eth.syncing = false, 16 peers
Cross-Platform Peer Discovery	Confirmed	Windows and WSL nodes detected each other as active peers
Official Infrastructure Connectivity	Confirmed	Multiple DAC Authority and RPC nodes connected on both nodes
Protocol Compatibility	Confirmed	eth/66, eth/67, eth/68, and snap/1 observed

Background

Following the previous infrastructure validation reports in this series, an additional verification cycle was performed after updating both environments to the latest node release. The test was designed to confirm that the Windows and Linux deployments continue to behave consistently under identical network conditions and continue to participate normally in the DAC Testnet after the update.

Objective

- Verify synchronization status on both nodes.
- Compare peer connectivity between the Windows and Linux (WSL) deployments.
- Validate cross-platform interoperability between the two node environments.
- Confirm continued connectivity to official DAC infrastructure (Authority and RPC nodes).
- Collect baseline observations for the planned Health Check Toolkit.

Test Environment

Node	Platform	Client	Result
Windows Node	Windows	gdacnode v1.11.6	eth.syncing = false, peers = 17
Linux Node	Ubuntu (WSL)	gdacnode v1.11.6	eth.syncing = false, peers = 16

Test Procedure

eth.syncing, net.peerCount, and admin.peers were executed on both nodes through IPC attachment. Results were compared to evaluate synchronization status, protocol negotiation, peer discovery, and connectivity to official DAC infrastructure.

Observation

- Both nodes returned eth.syncing = false.
- Windows maintained 17 peers, while Linux (WSL) maintained 16 peers.
- Both nodes detected each other as active peers.
- Multiple DAC Authority nodes were connected on both platforms.
- Multiple DAC RPC nodes were connected on both platforms.
- Protocol compatibility observed across eth/66, eth/67, eth/68, and snap/1.

Technical Findings

- 1 Full synchronization was achieved on both platforms.
- 2 A peer-count difference of one connection is considered normal under decentralized peer discovery.
- 3 Cross-platform connectivity (Windows ↔ Linux WSL) operated correctly.
- 4 Connectivity with official Authority and RPC infrastructure remained stable.
- 5 No indication of peer isolation, network partition, or synchronization failure was identified.

Technical Analysis

The validation demonstrates that the current DAC networking layer provides consistent behaviour across native Windows and Linux WSL deployments. Successful mutual peer discovery indicates that the two environments participate as independent network peers rather than isolated local instances. The observed peer-count variance is expected, since peer discovery is dynamic and continuously optimized by the networking layer — consistent with the peer-topology behaviour documented in earlier reports in this series.

Infrastructure Assessment

From an infrastructure perspective, these results increase confidence that independent operators can maintain multiple nodes across different operating systems without introducing synchronization inconsistencies. The validation also provides a reliable baseline for the planned automated Health Check Toolkit, where synchronization status, peer count, protocol compatibility, and infrastructure connectivity can be monitored automatically rather than checked manually via IPC.

Recommendations

- Integrate eth.syncing verification into the Health Check Toolkit.
- Include peer-count monitoring with configurable thresholds.
- Detect Authority/RPC connectivity automatically.
- Record protocol versions negotiated with peers.
- Add cross-node validation when multiple independent nodes are available.

Conclusion

Component	Status
Windows node synchronization	Fully synchronized — eth.syncing = false, 17 peers
Linux (WSL) node synchronization	Fully synchronized — eth.syncing = false, 16 peers
Cross-platform peer discovery	Confirmed — nodes detected each other as active peers
Official Authority / RPC connectivity	Stable on both platforms
Protocol compatibility	Confirmed — eth/66, eth/67, eth/68, snap/1
Overall infrastructure status	Healthy — no corrective action required

The latest validation confirms that both Windows and Linux (WSL) nodes are fully synchronized, maintain stable connectivity with the DAC Testnet, successfully discover each other, and communicate with official Authority and RPC infrastructure. No infrastructure issues requiring corrective action were identified during this validation cycle.